

Project summary

Application ID

111474-2026_FLF

Application Type

Future Leader Fellowship

Administering institution

Queensland University of Technology

Primary Applicant

Andrew Stephens

Years Post-PhD

7 to 10 years

Grant Subcategory

Level 2

Justification (if applying for a higher or lower level FLF)

Project title

A Non-Invasive Closed-Loop Cardiopulmonary Resuscitation Feedback Device

Lay title

A Smart CPR Device for Improving Patient Survival After Cardiac Arrest

Alignment with goals

Prevent, detect, and manage risk early

Research category

Biomedical, Clinical

Funding commencement date

1 - Jul

Project synopsis**Background**

Cardiac arrest is a sudden and life-threatening event that can happen unexpectedly. Good cardiac arrest outcomes rely on high-quality cardiopulmonary resuscitation (CPR) until a defibrillator can shock the heart back into rhythm. Current CPR guidelines assume a one-size-fits-all approach and may not provide the best support for all people in all environments. This project aims to develop a "smart" CPR device that can non-invasively measure brain oxygen levels and provide feedback to the CPR provider on how to improve their CPR technique to achieve the best outcomes for the patient.

Methodology

I will conduct co-design studies with paramedics to investigate the best ways to integrate the sensor into the existing cardiac arrest treatment workflow and how to best provide feedback to paramedics while delivering CPR. Co-design will be undertaken with key paramedic collaborators, and the resulting design will then be iteratively evaluated in working groups of paramedics and paramedicine trainees, who will provide feedback on the effectiveness and ease of use of the smart CPR device.

Next, clinical research will be conducted in two hospital ICUs to demonstrate the sensor's accuracy in a human population. The sensor will be trialled in patients treated in intensive care who are at risk of low-oxygen states. Non-invasive readings from the developed sensor will be compared with invasive blood-gas draws as the gold standard to demonstrate the device's accuracy.

Finally, an observational study will be conducted with two ambulance services to develop the CPR feedback algorithm. The validated sensor will be deployed during cardiac arrest calls and will collect data on brain oxygenation, CPR technique, long-term, and short-term patient outcomes. All of these data will be combined using control systems engineering to develop the CPR feedback system, which will give advice on compression techniques to optimise cardiac arrest outcomes.

Impact

This is the first device of its kind to provide personalised CPR feedback tailored to each patient's needs. The main outcome of this research will be a new device ready for testing in patients with cardiac arrest. This research will also generate new knowledge about the links between CPR, oxygenation in the brain and long-term outcomes for patients, including neurological disabilities.

Translation and Dissemination

Device translation will occur through a combination of spin-out from QUT, with an exit strategy that includes company buyout or licensing, partnering with existing defibrillator manufacturers
Dissemination will include publication in open-access, top-tier journals and conference presentations.

Research tags - categories

Biology, Condition, Information Technology, Treatment

Research tags - keywords

Brain and Nervous System (Biology), Arrhythmia (Condition), Personalised Treatment (Treatment), Guidelines and Decision Making Tools (Treatment), Devices (Treatment), CPR (cardiopulmonary resuscitation) (Treatment)

Ethics requirements

Human

Biosafety requirements

Research Benefit

☐ Advocacy ☐ Clinical Evidence ☒ Healthcare Programs ☐ Healthy Living ☐ First Nations Heart Health ☒ Commercialisation ☐ NA

Healthcare Programs

This research aligns strongly with the NHF programmes on cardiac arrest and CPR education and can link in with their advocacy for AED rollout in public spaces.

The goal is to integrate this technology into public defibrillators, enabling any bystander to deliver high-quality CPR without prior training. This approach would fundamentally change how CPR is delivered and taught globally. Creating a closed-loop feedback device could supersede the long-standing CPR guidelines, which have modest efficacy. Moving towards an individualised, dynamic treatment plan for each patient and setting could correct ineffective CPR delivery and reduce mistakes during prolonged CPR caused by fatigue.

Commercialisation

The goal is to translate this device into a commercial product. Starting as a QUT spin-out, we will continue device development and validation, building a body of evidence on its efficacy and cost-benefit.

We seek to partner with global defibrillator companies (company 1, company 2, company 3) to translate this research into existing commercial devices. Technology transfer will be achieved by licensing or IP sale.

CPR quality drives cardiac arrest outcomes and significantly affects long-term recovery and the occurrence of further cardiac events, aligning strongly with the Heart Foundation's 25-year vision goal to "manage heart disease and prevent further events".

Eligibility

No Tobacco Declaration

Confirm

AI Tools



Yes



No

AI Declaration



Yes

AI Declaration Information

Grammarly used to check spelling and grammar.

Citizenship Status

Australian Citizen

CIA Confirmation

Yes

Prior Relevant Research

Yes

Enrolment FTE

1.0

PhD Thesis Pass Date

18/02/2019

Post-Fellowship plans

My goals are:

1. Continue to grow my lab, driving research quality and impact
2. Instil in my students a passion for applied research; give them the skills and resources to translate their research into clinical practice
3. Continue to grow my international reputation as a leading researcher

This fellowship will contribute to these career plans by enabling me to pursue a research-focused academic position and maintain career momentum at this critical early- to mid-career juncture. Data obtained from this research will be leveraged for larger, focused clinical trials. This fellowship would recognise my contributions to cardiovascular engineering.

Project overview

Background and Preliminary Data

A proof-of-concept non-invasive oximeter device employing two light frequencies was manufactured and evaluated in healthy volunteers. The developed sensor was placed on their arm, and desaturation was caused by inflation and deflation of a blood pressure cuff. A clinical oximeter device was used as a comparison against the developed sensor, and a strong correlation was observed between the clinical oximeter and the developed sensor, providing initial confidence in the sensing mechanism. A second prototype has been developed using flexible wearable technologies.

Research Plan

1. Co-design study (Year 1)

A co-design study will investigate how compression feedback should be delivered to paramedics during CPR. Two experienced paramedics will help to define and refine potential feedback formats. Findings will inform three focus groups (4-8 paramedics each group) where participants will complete brief quantitative feedback forms rating clarity, usability, cognitive load, and perceived usefulness to identify the preferred feedback approach for field use.

2, Patient data from two hospital wards (Year 1-3)

The device will be evaluated in two adult intensive care units among patients receiving invasive mechanical ventilation for acute respiratory failure. Patient skin colour and age will be recorded, and the device will continuously monitor cerebral oxygenation in ICU. Readings will be matched against routine invasive blood draws. Deming regression modified Bland-Altman analysis will be used to assess precision and accuracy. Experiments will be conducted with 50 participants to achieve a 95% confidence interval (at 3%). Experimental data will be used to calibrate the sensor to account for

changes in skin pigmentation and age.

3. Observational ambulance trial and feedback algorithm development (Year 3 - 4)

An observational ambulance study will enrol 100 paramedic-attended cardiac arrest patients, with 10 prototype head-mounted devices distributed across (two major ambulance services in Australia).

Developed sensor readings will be compared with compression metrics from standard CPR pressure pads during resuscitation. Resulting data will be used to develop a CPR feedback algorithm, to be evaluated in future animal trials.

Impact

A CPR feedback device could fundamentally change CPR delivery, resulting in personalised therapy. Immediate knowledge gain about pre-CPR oxygenation levels (unwitnessed arrest) and short-term neurological outcomes.

Immediate knowledge gain about skin colour and oximetry accuracy.

Translation

An initial QUT spinout will be used to gather more data to drive value inflexion points for commercialisation (sensor accuracy, algorithm development, improved outcomes).

Dissemination of immediate knowledge gain will be done through OA publications and presentations.

Research environment

Research Environment and International Standing

QUT is ranked #1 in Australia and #16 globally for biomedical engineering (Shanghai Global Rankings, 2024). I have adjunct appointments at (two major cardiac hospitals in Australia), giving me access to two of Australia's largest cardiac hospitals for in-hospital trials. I have collaborated with (two major ambulance services) on this project, giving me access to two major paramedic services for the proposed observational trials.

Quality and Experience of Supervision: Redacted

Infrastructure: Key infrastructure provided by QUT for this project includes animal research facilities (QUT MERF), surface-mount and flexible-circuit manufacturing capabilities, PoC blood-gas analysers, and clinical trial support at all stages.

Available Funding: This research has been supported by QUT (\$200,000 confirmed between 2025-2027), with a further PhD scholarship and internal research funding committed if this fellowship is successful. This research has been supported by The Prince Charles Hospital Foundation (\$ some money) to run animal trials for the device (outside the scope of this fellowship).

Research plan and figures

Research plan

111474-2026_FLF_Stephens_Research_Plan.pdf
529.9 KB - 25/06/2026 10:59 PM

Total Files: 1

Expected outcomes

Expected Outcomes: The expected outcome of this research is a validated sensor and algorithm that provides a CPR provider with feedback on how to optimise the compression technique for better outcomes. The device feedback mechanism will be co-designed with paramedics to fit into existing workflows and deliver the information in out-of-hospital conditions. Device accuracy will be quantified with patients across two intensive care units. The feedback algorithm will be developed using observational data collected during ambulance-attended cardiac arrests, with the ultimate goal of incorporating this device into public AEDs.

Benefit to Cardiovascular Health: A haemodynamic-derived CPR feedback system could eventually allow even untrained bystanders to provide paramedic-quality CPR. By focusing on maximising oxygenated blood flow to the brain, the CPR system could improve survival and reduce the likelihood of negative neurological outcomes associated with prolonged arrest.

Next Steps: This work is expected to lead to larger, more rigorous studies. The data collected during this study will be used to attract industry partners and leverage funding to continue development. Device translation will be supported by industry investment or through formal partnerships funded by the CRC or other industry-based funding schemes, with future research focusing on the clinical improvements.

Translation and dissemination

End-Users

1. Hospital arrest responders
2. Paramedics
3. Bystander/witness arrest responders

As part of the co-design methodology, end-users (paramedics) will be closely involved in the development and evaluation of the device. Outcomes from the research will be presented through publications, presentations, and seminars delivered to paramedics. Knowledge gained from understanding pre-CPR cerebral oxygenation and neurological outcomes can inform arrest guidelines on resuscitation futility.

Clinical Translation and Impact

This device has undergone a medtech commercialisation advancement program (MCAP) at the Australian Medtech Manufacturing Centre, funded by Monash University and the Victorian State Government. The MCAP program outlined a structured pathway to commercialisation, serving as a blueprint for clinical translation. The CPR device will first be translated into mechanical chest compressors (CPR vests) to ensure that the CPR feedback is followed exactly and demonstrate improved patient outcomes. Next, the device will be translated into paramedic-delivered CPR, as they can more easily follow the CPR feedback guidelines. Finally, the device will be incorporated into publicly available defibrillators, enabling any bystander to provide paramedic-quality CPR to cardiac arrest victims. Ultimately, bystander-enabled technology would be achieved by partnering with major medical device companies and integrating our technology into their public defibrillators (Zoll, CardioHelp, Philips, to name a few).

The impact of this research is that CPR would shift from a one-size-fits-none approach to a personalised therapy that can be delivered independent of provider skill and experience. This research aligns with the NHF vision goal 4 - "manage heart disease effectively and prevent further events". Tailoring CPR to each individual will ultimately lead to better long-term outcomes, reducing the incidence of neurological impairment, the likelihood of rehospitalisation due to post-arrest heart failure, and improving post-arrest patient independence.

\$49,143.00	\$49,618.00	\$49,378.00	\$49,937.00	\$198,076.00
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Other funding

Previous Funding

Yes

Previous Funding Details

This research was previously funded by Monash University (\$some money; 2019-2023) to develop initial prototypes and to perform proof-of-concept and safety testing of the CPR feedback device, which was successfully completed. Monash also paid for the device to undergo a commercial viability review with an independent consultant (valued at approx. \$some money). The work done at Monash University identified several areas for improvement, which are currently being addressed through my fellowship with The Prince Charles Hospital Foundation (2024-2027). The work supported by the Prince Charles Hospital Foundation has allowed us to update the device's design and create an "ambulance-ready" system. My current fellowship work includes human trials (evaluating device safety), animal trials (evaluating sensor accuracy), and a small number of pilot ICU trials (N = 20; proof of concept in a representative human population).

This proposed research program advances that work by optimising the user interface and feedback mechanism in paramedic co-design trials, building on the pilot work conducted in an ICU population, and conducting a large-scale observational trial to develop the feedback algorithm. All necessary steps towards clinical translation and commercialisation.

Fellowship Funding

Yes

Fellowship funding details

I will apply for a fellowship for this project through an NHMRC Investigator Grant EL2, valued at \$327,360 per year, commencing in 2028 (there is a one-year lag between NHMRC rounds). NHMRC fellowship applications will close before the NHF fellowship outcomes are announced.

Top Publications

#	Type	Date	Work Title	Journal	Contributors
001	journal-article	2023	ECMO PAL: using deep neural networks for survival prediction in venoarterial extracorporeal membrane oxygenation	Intensive Care Medicine	Stephens, A.F., Šeman, M., Diehl, A., Pilcher, D., Barbaro, R.P., Brodie, D., Pellegrino, V., Kaye, D.M., Gregory, S.D., Hodgson, C.
002	edited-book	2025	Mechanical Circulatory and Respiratory Support		
003	journal-article	2020-07-01	Coping with COVID-19	Anaesthesia	A. L. Clarke, A. F. Stephens, S. Liao, T. J. Byrne, S. D. Gregory
004	book-chapter	2025	Preclinical evaluation	Mechanical Circulatory and Respiratory Support	Andrew F. Stephens, Lisa Prah Wittberg, Steven Jacobs, Jan Roggenkamp, Jutta Arens, Libera Fresiello
005	journal-article	2019-08-15	A novel fibre Bragg grating pressure sensor for rotary ventricular assist devices	Sensors and Actuators, A: Physical	Andrew F. Stephens, Andrew Busch, Robert F. Salamonsen, Shaun D. Gregory, Geoffrey D. Tansley
006	journal-article	2022-09	Ex vivo evaluation of personal protective equipment in hands-on defibrillation	Resuscitation Plus	Andrew F. Stephens, Michael Šeman, Ziad Nehme, Aleksandr Voskoboinik, Karen Smith, Shaun D. Gregory, Dion Stub
007	journal-article	2020-06-01	An advanced mock circulation loop for in vitro cardiovascular device evaluation	Artificial Organs	Shaun D. Gregory, Jo P. Pauls, Eric L. Wu, Andrew Stephens, Ulrich Steinseifer, Geoff Tansley, John F. Fraser
008	journal-article	2021-05	Rotary Ventricular Assist Device Control with a Fiber Bragg Grating Pressure Sensor	IEEE Transactions on Control Systems Technology	Andrew F. Stephens, Andrew Busch, Robert F. Salamonsen, Shaun D. Gregory, Geoffrey D. Tansley
009	journal-article	2021-06-01	Comparison of Circulatory Unloading Techniques for Venoarterial Extracorporeal Membrane Oxygenation	ASAIO Journal	Andrew F. Stephens, Devindi Wanigasekara, Vincent A. Pellegrino, Aidan J.C. Burrell, Silvana F. Marasco, David M. Kaye, Ulrich Steinseifer, Shaun D. Gregory
010	journal-article	2022-07-01	Fabrication and Characterization of Tissue-Mimicking Phantoms for Ultrasound-Guided Cannulation Training	ASAIO Journal	Rezan Jafary, Sophie Armstrong, Timothy Byrne, Andrew Stephens, Vincent Pellegrino, Shaun D. Gregory

Justification - Publications**Publication #1 (First Author)**

ECMO PAL: using deep neural networks for survival prediction in venoarterial extracorporeal membrane oxygenation.

Contribution to Field: This work was the first exploration of ECMO outcomes using big data and artificial intelligence techniques and was published in Intensive Care Medicine (IF 41.8 at the time). Using AI, I developed a tool for predicting hospital survival in patients receiving artificial heart-lung support for cardiac indications (VA ECMO). The tool ECMO PAL VA is now hosted online and freely available at ecmo-pal.icu. The AI tool outperformed the current gold-standard risk adjustment scores by a significant margin (over 10%). I am currently involved in an international working group through ELSO to incorporate our AI tool into global risk adjustment, enabling 660 member hospitals to benchmark their risk-adjusted centre performance against similar centres. The article is in the top 5% of all articles published of a similar age on Altmetric.

My Role: I led an international team of clinicians and engineers to design the study, develop the AI, build the web app, publish the manuscript, and integrate the study into ELSO systems.

Publication #2 (Second Editor)

Textbook of Mechanical Circulatory and Respiratory Support (2nd Edition), Elsevier

Contribution to the Field: This is the most comprehensive textbook on mechanical circulatory support (MCS). Consisting of 86 expert authors, the textbook covers topics ranging from basic pathophysiology and MCS implantation to advanced patient management, device design, regulation, pathways to market, and post-market monitoring. The textbook is used as a standard teaching text at at least one university (the University of Malaya) and was commercially successful enough for Elsevier to request a second edition.

My Role: I coordinated all contributing authors and reviewed all chapters in Elsevier's book-writing tool, "ELSA". I regularly coordinated with the publisher and provided instructions and support to the authors. I wrote one chapter as first author (Publication #4) and co-wrote two chapters as second author (not in this list).

Publication #3 (Second Author)

Coping with COVID-19: Ventilator Splitting with Differential Driving Pressures Using Standard Hospital Equipment

Contribution to Field: This was the second article published on ventilator splitting at the height of the first 2020 COVID-19 wave. The article explores ventilator splitting using standard hospital equipment as an alternative to unregulated 3D-printed parts. The article ranks in the top 0.4% for research prominence (SciVal) and has been featured in dozens of Australian and international news outlets. This work is ranked in the top 5% of all research articles (all-time, Altmetric). The work led to my involvement in the now TGA-approved GeVentor (gekkos.com/gekko-medical) and to discussions with the British High Commission about ventilator splitting in the UK during its first wave.

My Role: I designed and conducted the experiments, analysed the data, and wrote the first draft of the manuscript.

Publication #4 (First Author)

Preclinical Evaluation. Chapter in Textbook of Mechanical Circulatory and Respiratory Support

Contribution to Field: This book chapter outlines common methods for evaluating mechanical circulatory support devices, ranging from in silico optimisation to preclinical animal models. The chapter is used as a part of the teaching syllabus at the University of Malaya for their unit "Mechanical Circulatory and Respiratory Support".

My Role: I led the writing of this paper, collaborating with several international experts in Preclinical evaluation methods.

Publication #5 (First Author)

A Novel Fibre Bragg Grating Pressure Sensor for Rotary Ventricular Assist Devices

Contribution to Field: This paper describes the first fibre-optic pressure sensor designed specifically for ventricular assist devices. The fibre-optic pressure sensor has benefits over electrical sensors due to its high sensitivity and resistance to biological ingress. This work attracted commercial interest from two artificial heart device companies and has resulted in the grant of one patent, on which I am a co-inventor.

My Role: I was co-inventor of the technology and designed and executed the study.

Publication #6 (First Author)

Ex Vivo Evaluation of Personal Protective Equipment in Hands-On Defibrillation (First Author)

Contribution to Field: This work was conducted as basic science prior to a clinical trial for hands-on defibrillation across Victoria (AUGMENT-VA). The work investigated the electrical safety of commonly used personal protective equipment (medical gloves, rescuer trousers) and found it to be safe when used correctly. This controversial work has already begun to change local attitudes towards the safety of hands-on defibrillation, which reduces harmful chest compression interruptions. The research formed the basis of my testimony to WorkSafe Victoria, which informed their decision to allow the trial to go ahead.

My Role: This work was conducted in collaboration with Ambulance Victoria and the Alfred Hospital. As the only electronic engineer on the team, I designed and conducted the experiments, interpreted the results, and wrote the manuscript.

Publication #7 (Middle Author)

An Advanced Mock Circulation Loop for In Vitro Cardiovascular Device Evaluation (Middle)

Contribution to Field: This manuscript outlines the most sophisticated benchtop cardiovascular simulator (mock loop) to date. This work forms the basis for many mock loops developed around the globe, including one that I helped build for Cook Medical. This work has shaped the pre-clinical evaluation of mechanical circulatory support devices globally.

My Role: I developed aspects of the mock circulation loop, including automatic test beds, the Frank-Starling mechanism, and fine-tuning of the automated control loops. I wrote the corresponding sections of the manuscript.

Publication #8 (First Author)

Rotary Ventricular Assist Device Control with a Fibre Bragg Grating Pressure Sensor

Contribution to Field: This work combines my previous work in biomedical sensors (Pub #5) and a smart control system to create a fully realised closed-loop control system for artificial heart devices. The developed control system expanded beyond the Frank-Starling mechanism to replicate the sympathetic and parasympathetic nervous system, adapting to a broader range of cardiovascular states, notably heavier exercise. The work resulted in the granting of a second patent, on which I was co-inventor.

My Role: I led all aspects of the research.

Publication #9 (First Author)

Comparison of Circulatory Unloading Techniques for Venoarterial Extracorporeal Membrane Oxygenation

Contribution to Field: This is the only study to compare different circulatory unloading techniques for ECMO and quantify their effects on patient haemodynamics and their capacity to reduce pulmonary oedema. The work was done in collaboration with the Alfred Hospital, and its recommendations were immediately adopted clinically within the hospital. Upon release, the work was retweeted over 50 times and ranked in the top 3% of articles of similar age on Altmetric, demonstrating high clinical interest.

My Role: I designed, led, and executed the study. I also collaborated with clinicians to interpret the study results.

Publication #1 from "Updated Track Record Form" (First Author)

ECMO PAL VV: Using Deep Neural Networks for Survival Prognostication in Venovenous Extracorporeal Membrane Oxygenation

Contribution to Field: This research expands on Publication #1 by extending it to patients receiving artificial heart-lung support for respiratory collapse (VV ECMO). Along with Publication #1, the work is in the late stages of rollout into the ELSO risk adjustment processes, allowing over 660 centres access to use it for risk-adjusted benchmarking purposes.

My Role: I led the international, interdisciplinary team in developing and refining the AI models and am overseeing their rollout and implementation.

Top Awards and Distinctions

#	Type	Date	Organisation	Department	Title
1	Distinction	2017	University of Queensland		Medical Innovators of the Year
2	Distinction	2017	Griffith University		Outstanding Contribution to Teaching
3	Distinction	2016	CRE ACTIONS		Young Investigator Award - Runner Up
4	Distinction	2023	Extracorporeal Life Support Organization		Best Conference Presentation

Justification - Awards and Distinctions

#1 - I was co-awarded the Medical Innovators of the Year award by UQ for our work on The OpenHeart Project, an open-source repository of information and masterclasses on mechanical circulatory support. That project has since become defunct, but led directly to my involvement in the Textbook of Mechanical Circulatory and Respiratory Support (Publication #2).

#2 - I was awarded Outstanding Contribution to Teaching in 2017 by Griffith University for my sessional work in the first-year unit "Programming for Engineers". Previously, in 2016, I was also nominated.

#3 - As a PhD student, I received the runner-up award for my presentation at the CRE ACTIONS Young Investigator Research Symposium. Judging was conducted by a panel of international experts in the field who contributed to the CRE.

#4 - Along with my co-author (Michael Seman), we won the best conference presentation at the ELSO conference for our work outlined in Publication #1.

A. Research Plan

Aim: This program aims to develop and validate a closed-loop cardiopulmonary resuscitation (CPR) feedback device that gives real-time feedback to a CPR provider to optimise the amount of oxygenated blood delivered to the brain. The goal is to develop a “smart” device that can be integrated into public electronic defibrillators and enable even untrained bystanders to provide optimal CPR for any patient in any situation. The research program is broken down into three studies (Figure 1):

- 1) A co-design study with paramedics to investigate the best way to integrate the sensor into the existing cardiac arrest workflow and the best feedback mechanism for providing compression adjustment advice during cardiac arrest.
- 2) A clinical trial in two intensive care units to demonstrate the sensor’s accuracy in an intensive care population
- 3) An observational study conducted in ambulances to collect sensor data during cardiac arrest calls. This data will be used to develop the CPR feedback algorithm.

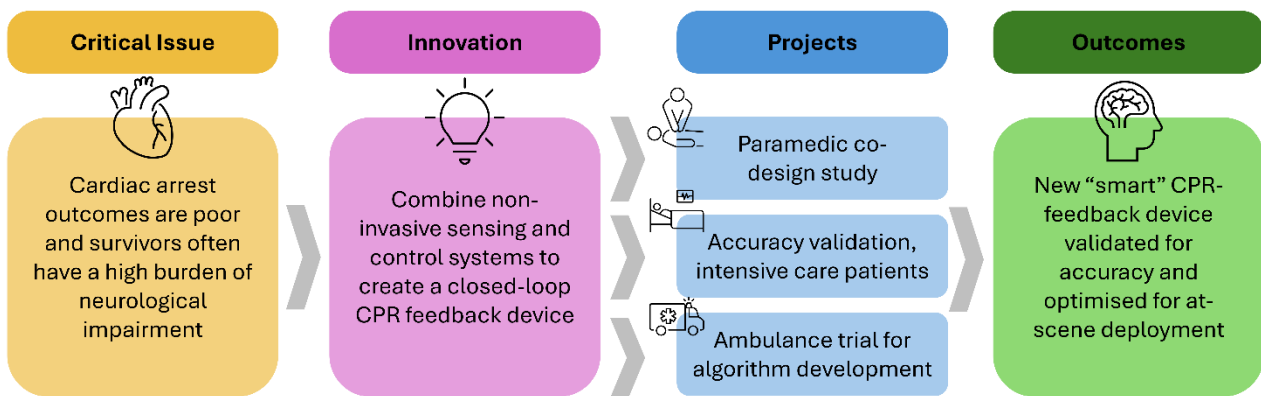


Figure 1: The overall program plan includes a co-design trial and two major clinical investigations

Background: Out-of-hospital cardiac arrests account for nearly 26,000 emergency callouts each year in Australia, with a survival rate of 8-10%, even with paramedic-delivered CPR (1,2). Cardiac arrest costs the Australian healthcare system approximately \$2 billion annually, with productivity losses comparable to all cancers combined (3). Good cardiac arrest outcomes depend on high-quality CPR and early defibrillation (4). Current CPR guidelines are based on only modest clinical evidence, and there is a demonstrated upper limit on guideline adherence and improvements in patient outcomes (5,6). On the other hand, cerebral oxygenation is an independent indicator of patient outcomes, and its measurement in real-time can facilitate personalised CPR based on individual patient needs (7,8).

This research will develop CerebrOx, a closed-loop CPR feedback system based on measured cerebral oxygenation. CerebrOx uses near-infrared spectroscopy (NIRS) to non-invasively measure oxygenated and deoxygenated haemoglobin in blood, as well as other chromophores such as melanin in the skin (9). Specific wavelengths of light are carefully chosen to pass through the skin and bone while being scattered by the chromophore of

interest. Five wavelengths have been chosen to allow the device to be calibrated for different skin colours.

Preliminary Data

Previously, a CerebrOx prototype was developed using initial seed funding and fellowship support from Monash University and The Prince Charles Hospital Foundation. Initial devices were developed on a single rigid board employing two wavelengths of light (one for oxygenated and one for deoxygenated haemoglobin). These initial prototypes were tested in two healthy volunteers and compared with a commercial oximeter, showing a strong correlation but an inability to quantitatively measure blood oxygen concentration due to limitations of the modified Beer-Lambert algorithm employed (10). Additionally, the initial prototype was found to exceed recommended operating temperatures (IEC 60601-1) due to the light-emitting diode drivers.

Since 2024, the prototype has been updated to include a flexible “consumable” sensor and stand-alone driver unit (Figure 2). The updated device also incorporates five wavelengths of light ($\lambda = 690, 730, 770, 810, \text{ and } 850 \text{ nm}$) to enable quantitative measurements of regional oxygen saturation and compensate for different skin colours. This updated prototype is currently being evaluated in healthy human trials and animal experiments, which are expected to finish by the end of 2026. Animal experiments are being conducted in landrace pigs using a cardiac arrest model to evaluate sensor accuracy against gold-standard invasive measurements across a wide range of saturation states. So far, one trial has been completed with five more scheduled. The animal underwent initial cardiac arrest for five minutes (low saturation), followed by attempted resuscitation (mechanical chest compressions, adrenaline, and shocks) for twenty minutes (moderate saturation), and finally, initiation of extracorporeal life support for one hour (high saturation).

Preliminary results from this experiment showed qualitative tracking of oxygenated and deoxygenated haemoglobin at each of the stages of support (Figure 3). Correlation analysis against invasive arterial blood-gas monitoring taken throughout the procedure is still ongoing. Data from the animal trials will be corroborated by data collected from healthy volunteers using the improved prototype, with healthy volunteer recruitment currently underway. In a reactive ischaemic model using forearm tissue, the developed CPR sensor is being correlated with an off-the-shelf oximeter to produce quantitative measurements of regional oxygen saturation. Although still ongoing, preliminary data from these experiments have been used to inform the methodology proposed below, building on the experimental complexity and moving towards clinical realism. The proposed methodological approach is broken into three studies.



Figure 2: New flexible CerebrOx Sensor and control box. Old rigid CerebrOx sensor prototype (green board, bottom right).

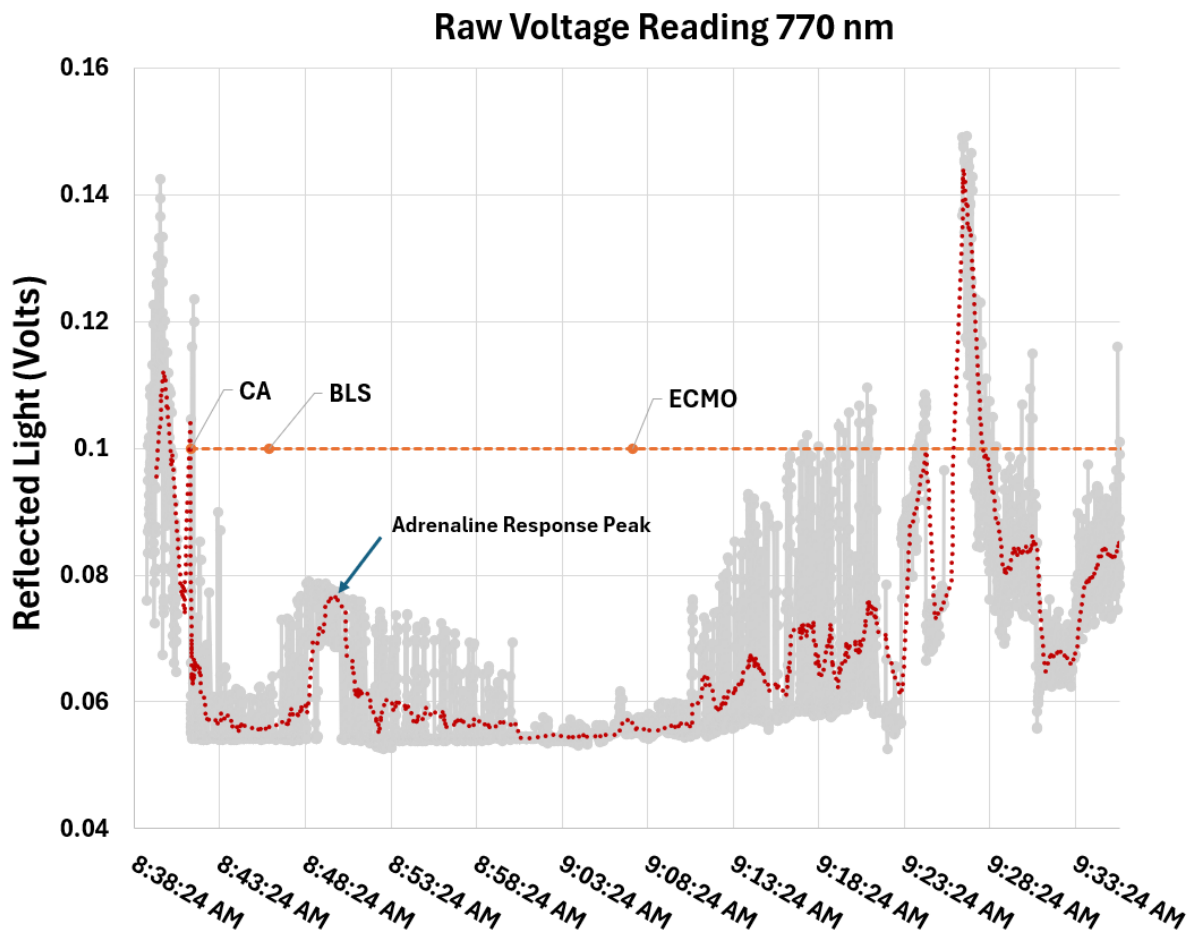


Figure 3: Preliminary data from a landrace pig shows the moving average of the reflected 770 nm wavelength tracking changes in cerebral oxygenation during cardiac arrest and subsequent resuscitation experiments. Note: this is preliminary data and analysis is still underway.

CA- cardiac arrest, BLS – basic life support, ECMO – extracorporeal membrane oxygenation.

Methods

Study 1: Feedback Mechanism Paramedic Co-Design

To maximise the potential for clinical translation, it is vital to ensure the device integrates seamlessly with the existing clinical workflow. It is also crucial to design a feedback mechanism that promotes adherence to the CPR algorithm's instructions. *Study 1* will conduct co-design experiments with paramedics to optimise CerebrOx integration and the feedback algorithm into the existing cardiac arrest treatment workflow. Co-design studies will be conducted in collaboration with (two major ambulance services in Australia), leveraging existing research collaborations. Five co-design studies will be conducted at each site, with four to six participants per study (N = 40 - 60). Chosen numbers are based on estimates of research-interested members of each site.

We will work with experts in the QUT School of Design to develop three feedback mechanisms consisting of audio, visual, and haptic cues, as well as pictographs. Each feedback mechanism will be implemented into the device, which will have a series of pre-programmed instructions to create different CPR scenarios. Each participant will provide 10

minutes of CPR on a medical mannequin using all three feedback mechanisms, following different compression instruction sets. Participants will provide quantitative and qualitative scores for each feedback mechanism, as well as for overall device usage, rating the clarity of instructions, cognitive load, suitability for expected environments, and integration with workflow.

Qualitative scores will be aggregated and presented as box-and-whisker plots for each evaluated category. Statistical significance for each category will first be assessed using the Friedman test for ordinal data, followed by pairwise Wilcoxon Signed-Rank tests. Qualitative data will be analysed for common themes and incorporated directly into the development of the feedback mechanism for future experiments.

The work will take place over the first year and part of the second year of the fellowship. The expected outcome of this study is a co-designed CPR feedback mechanism that integrates best into the existing workflow and provides clear and consistent instructions to the CPR providers.

Study 2: Sensor Accuracy in an Adult Intensive Care Population

The sensor is currently being evaluated in animal models of cardiac arrest, but has not been validated in a human cohort in a clinical setting. This study aims to demonstrate sensor accuracy across two intensive care units, leveraging existing research partnerships. CerebrOx will be evaluated in adult intensive care patients receiving invasive mechanical ventilation due to acute respiratory failure ($\text{PaO}_2/\text{FiO}_2$ ratio < 200 mmHg). Minimal patient demographics will be collected, only skin colour (as determined by the attending clinician) and age, both necessary to analyse sensor results.

CerebrOx will continuously read cerebral oxygenation readings for 24 hours after patient admission. These readings will be matched against arterial and mixed-venous PO_2 and PCO_2 , which will be drawn as part of routine care throughout the intensive care stay, with additional blood draws performed during any unexpected periods of desaturation. Continuous non-invasive measures of oxygenation will be time-matched and compared with blood draws as the gold standard. Deming regression and bootstrapping will be used to produce a modified Bland-Altman analysis, providing precision and accuracy stratified by skin colour (11)(12). Skin colour and age will be explored as sensor effects using linear mixed-effects models, investigating both device accuracy (error) and absolute differences in device readings (bias). Experiments will be conducted on 30 participants per site ($N = 60$) based on a 95% confidence interval of accuracy within 3% of the gold standard (minimum 47 patients). If recruitment goes well, additional patients or sites may be requested to increase confidence. Data from these experiments will be used to apply calibrations to the sensor and developed feedback mechanism (via generalised linear modelling or deep neural network regression) which will account for changes in skin pigmentation and age.

These experiments will be conducted over years one to three of the program, with the majority of ethics and governance work completed in parallel with Study 1 during the first year of the fellowship. The outcome of this study will be the validation of CerebrOx in a pathophysiological human cohort, as well as knowledge gained about skin colour and age and their effects on NIRS readings (forming the basis for publications).

Study 3: Algorithm Development using Prospective Observational Ambulance Data

Substantial clinical CPR data is required to develop the CPR feedback algorithm. To date, the development of CerebrOx has focused on quantifying the accuracy of the oxygen-sensing mechanism. This study aims to gather data that correlates the CPR technique (depth, rate, release) with cerebral oxygenation during CPR. Secondary analyses will aim to quantify NIRS measures to mortality and long-term outcomes. Leveraging the data obtained in Studies 1 & 2, this research will be a prospective observational trial conducted across two major ambulance services, building on the relationships strengthened by the co-design research outlined in Study 1.

Twenty CerebrOx devices will be deployed in intensive care ambulances (10 in each state), along with disposable sensors for each patient. The CPR sensor will be placed on the heads of patients receiving paramedic-attended CPR and will continuously collect cerebral oxygenation data throughout the resuscitation process. Simultaneously, compression data (depth, rate, release) will be collected from CPR pressure pads currently used by the ambulance services (Stat-Padz Plus, Zoll Medical Corp., MA, USA). CPR pressure data will be used to quantify CPR technique against beat-to-beat oximetry readings. Measured variables include cerebral oxygenation, compression depth, rate, release, arrest downtime, outcome (resuscitation vs non-survival), and neurological status at one-year follow-up (leveraging existing follow-up reporting within those services). There will be no other changes to clinical care.

Initially, we aim to include 100 enrolments over two years (50 per site). Despite expected enrolment difficulties, it should be achievable given the number of cardiac arrest callouts each year in each state, over 7,000 per year in Vic (13) and over 6,000 per year in Qld (14). Time-matched oxygenation and CPR compression data from the observational trials will be used to generate compression feedback algorithms, exploring traditional control methods (fuzzy logic, Kalman filter) and artificial intelligence (recurrent neural networks, long- and short-term memory networks). The controller objective will seek to optimise cerebral oxygenation using the compression technique as inputs (depth, rate, release). It will include a modelling pipeline that allows for automatic algorithm improvement as more patient data is fed into the model.

Three secondary investigations will also be conducted during these trials, focusing on (1) cerebral oxygenation in return of spontaneous circulation, (2) correlating cerebral oxygenation and long-term neurological outcome to derive oximetry-based cut-offs for CPR futility, and (3) adherence to current CPR guidelines and effects on cerebral oxygenation in relation to the return of spontaneous circulation. Each secondary investigation will leverage the data collected as described above and will be published to expand knowledge about cerebral oxygenation during cardiac arrest.

Governance and management for this study will begin towards the end of the second year of the fellowship, with the observational trial conducted over the third and fourth years. If recruitment goes well, there is potential to increase the number of recruited patients in the final year. The expected outcome of this study is a novel control algorithm that can advise a CPR provider on adjusting their compression technique to optimise cerebral oxygenation.

RESEARCH PLAN – FUTURE LEADER FELLOWSHIP COMMENCING 2027

Application ID: 111474-2026_FLF

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Significance, Expected Outcomes, and Future Directions

This work aims to develop the first-ever haemodynamically-guided closed-loop CPR feedback device. A data-driven approach to CPR delivery will provide a personalised approach to life support, which is suited for each patient in each environment. By focusing on cerebral oxygenation optimization, the intent is to move beyond survival and begin aiming for stronger long-term outcomes, free of neurological impairment and dependence. Building on several years of sensor development work, the expected outcome of this fellowship is a validated, non-invasive cerebral oxygen sensor that can be paired with a CPR control algorithm developed using real-world observational data. The sensor will be co-designed with paramedics to ensure the best feedback mechanism and minimal disruption to the existing clinical workflow.

The data obtained in this fellowship will be used to leverage project development grants and industry investment, with the next steps being to test the CPR feedback algorithm in animal studies, before moving to human device trials and clinical translation. The CerebrOx device will remain in the research domain for as long as is practical, before forming the basis of a QUT spin-out company (at the time of patenting). The preferred exit strategy is to license or transfer the CerebrOx intellectual property prior to large-scale clinical trials for regulatory approval. Our preferred partner would be a large multinational defibrillator manufacturer company such as (several prominent manufacturers), as this would allow integration with public defibrillator technologies that would enable bystander use.

B. Project Milestones

Milestone Description	Expected Completion Date	Comments
Co-design ethics and governance completed	04/27	Year 1
First co-design work group completed	05/27	Year 1
ICU 1 ethics and governance completed	11/27	Year 1
Co-design studies completed	12/27	Year 1 reporting begins
First ICU patient enrolment	03/28	Year 2
30 ICU participants enrolled	08/28	Year 2
Ambulance trial governance granted (site 1)	12/28	Year 2
60 ICU participants enrolled	02/29	Year 3 Review data, decide on whether more enrolment is required
Ambulance trial governance completed (site 2)	03/29	Year 3
First ambulance trial patient enrolled	06/29	Year 3
50 ambulance trial patients enrolled	12/29	Year 3

Initial model architecture developed	03/30	Year 4 begins leveraging funding for further studies
100 ambulance trial patients enrolled	09/30	Review Year 4 review data and decide whether more enrolment is required. If not, finalise and reporting

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2024 Future Leader Fellowship

Application ID:

111474-2026_FLF

Applicant Name:

Andrew Stephens

Interviewee track record update

Please note: Only include details updated since the date the application was submitted, using the headings below (remove if not applicable).

Publication details:

- **Stephens A.F**, Seman M, Hackwill R, Deihl A, Pilcher D, Barbaro R.P, Brodie D, Pellegrino V, Kaye D.M, Gregory S.D, Hodgson C.L, ECMO PAL VV: Using Deep Neural Networks for Survival Prognostication in Venovenous Extracorporeal Membrane Oxygenation, *Critical Care*, 2026, (DOI: 10.1186/s13054-026-06032-7)

This publication continues my artificial heart-lung outcomes prediction work by using AI to predict outcomes for patients receiving support for lung and respiratory failure. The work was published in the 4th highest-rated journal in critical care medicine.